

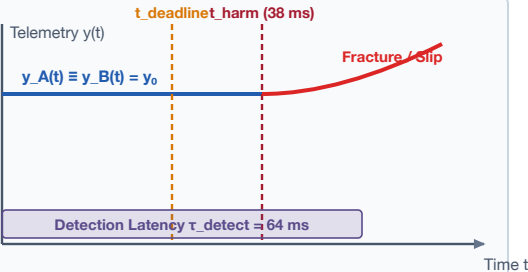
# THE DUAL EPISTEMIC BOUNDARIES: DETECTOR DEADLINES & EXPOSURE WALLS

Why Empirical Testing Alone Cannot Settle Open-World Safety, Demanding Architectural Authority Limits and Closure Tests

## 1. OBSERVABILITY DEADLINE LIMIT

### Identical Pre-Harm Telemetry

States  $S_A$  (Safe) vs  $S_B$  (Hazard) indistinguishable



- State  $S_A$ : Nominal surface ( $\mu \geq 0.60$ ) / sound core
- State  $S_B$ : Oil slick ( $\mu = 0.12$ ) / internal void
- Pre-Contact: Zero optical or tactile signature
- $\tau_{\text{detect}} > t_{\text{harm}}$ : Contact fracture occurs before reflex can decelerate jaw (38 ms)
- Theorem: When  $y_A(t) \equiv y_B(t) \forall t \leq t_{\text{deadline}}$ , safe intervention cannot exceed  $t_{\text{detect}}$

## 2. ASTRONOMICAL EXPOSURE WALL

### $n \geq \ln(1/\alpha) / p$ at 95% Confidence

Required failure-free hours grows reciprocally with  $1/p$

$p \leq 10^{-2}/h$  (Consumer App)

Time: 12.5 days

$n \geq 300$  h

Tractable (Unit Test)

$p \leq 10^{-4}/h$  (Industrial Cobot)

Time: 3.4 years

$n \geq 3.0 \times 10^4$  h

Affordable (Bench Rig)

$p \leq 10^{-6}/h$  (Autonomous Transit)

Time: 342 years

$n \geq 3.0 \times 10^6$  h

Fleet Limit (Correlated)

$p \leq 10^{-9}/h$  (SIL-4 / ASIL-D)

Time: 342,000 yrs

$n \geq 3.0 \times 10^9$  h

LOGICALLY INSUFFICIENT

### WHY FLEET SAMPLING FAILS AT THE TAIL:

1. Correlated updates & identical software reset trial clock
2. Safety drivers censor crashes  $\rightarrow$  missing counterfactuals
3. Simulators test only pre-programmed physics disturbances
4. Conclusion: Ultra-reliability demands structural containment!

## 3. RESIDUAL-CLAIMS REGISTER

### Converting Caveats to Engineering Work

4 Mandatory Fields for Every Open Assumption

#### FIELD 1: UNSUPPORTED CLAIM

e.g. Brake distance  $\leq 0.49$  m on concrete  
Missing Observable: Surface friction  $\mu < 0.15$

#### FIELD 2: OPERATIONAL CONTAINMENT

Structural velocity clamp  $v \leq 0.9$  m/s  
Kinetic energy  $E_k$  reduced 83% (1089 J  $\rightarrow$  182 J)

#### FIELD 3: ACCOUNTABLE OWNER

Named Lead Drivetrain Engineer owns parameter lock;  
Accepts operational throughput penalty

#### FIELD 4: FALSIFIABLE CLOSURE TEST

Integrate polarimetric sensor + 1 kHz slip reflex;  
Validated across  $N=1000$  wet floor trials

### FALSIFICATION & VERDICT GATE

Relocated Assumption?  $\rightarrow$  REJECT  
Independent Evidence?  $\rightarrow$  RETIRE GAP & LIFT CLAMP